

PACT: Pairwise-Anchored Calibrated Tuning for Single-Token Typed Decisions

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Abstract—Single-token typed-decision models answer a schema question by reading the logits of a few one-letter answer codes at a single position: they are fast and return a probability for every allowed answer, but they are trained with plain cross-entropy that ignores most of the structure in their training data. We study such a model whose data is curated as *contrastive pairs*—two contexts that differ in one edited fact that flips the answer—each carrying a machine-checked certificate that deleting the decisive sentence makes the fact unknown. We propose PACT, which turns this structure into four training terms that need no new annotation: a difference-in-differences margin over each pair that is invariant to any shared logit offset, a permutation-consistency term against answer-code position bias, an evidence-necessity term on certificate-verified ablated contexts, and an ordinal transport cost for rubric fields, plus a three-parameter contextual temperature. On a frozen 324-item holdout with three seeds, PACT matches the published recipe in accuracy (84.6 vs. 85.2%; McNemar $p \geq 0.50$ at every seed) while giving the lowest position bias of all runs (answer flips under relabelling 9.8 vs. 13.8%) and the lowest ordinal error on rubric fields (MAE 0.232 vs. 0.311). Against a control with the same optimiser and schedule but cross-entropy only, PACT is significantly more accurate at two of three seeds, halves the seed-to-seed spread and lowers NLL by 26%. Seed-matched ablations and pre-specified falsification tests locate these gains precisely: no single term raises raw accuracy, and the method’s value lies in robustness and stability rather than headline accuracy. Code, data splits, trained adapters and all run records are available at <https://github.com/BennyLinntu/PACT-Pairwise-Anchored-Calibrated-Tuning-for-Single-Token-Typed-Decisions>.

Index Terms—large language models, typed decisions, contrastive data, counterfactual supervision, position bias, calibration, parameter-efficient fine-tuning

I. INTRODUCTION

A *typed decision* is a classification whose label space is supplied at inference time by a schema: pick one of five outcomes, answer a true/false question, or place a case on a five-level rubric. Such decisions can be served efficiently by rendering the context and schema once, assigning each allowed answer a one-token code, reading the logits of those codes at one position, and normalising over them [1], [2]. There is no generated JSON to parse and no sampling, and every allowed answer receives a probability.

The training side of this interface is less settled. The open Nimble recipe [2] fine-tunes a 9B-parameter base model with LoRA [3] on 2,676 curated examples using cross-entropy over the candidate logits. Its curation, however, is far richer than

its objective. Every example belongs to a *contrastive pair* produced by editing at most eight words of one evidence sentence so that the reference answer flips, and every pair carries a certificate that records the decisive fact and verifies that removing either focus sentence leaves that fact *unknown*. The objective sees each example on its own: the pairing, the certificate and the arbitrariness of the letter assigned to each answer never enter the loss.

This paper asks what a training objective looks like when it does use that structure. Our premise is that the most informative supervision in contrastive data lies not in either example but in the *difference* between them—and in the counterfactual absence of the evidence—neither of which cross-entropy can express. We make three contributions.

- 1) **A structure-derived objective (PACT).** Four terms that use only structure already present in the data (Section IV): a difference-in-differences margin whose value is provably unchanged by any logit offset the pair shares (Proposition 1); a Jensen–Shannon consistency term across answer-code relabellings; an evidence-necessity term on certificate-verified ablated contexts, which the released recipe discards; and a squared earth-mover cost for ordinal rubric fields. A three-parameter contextual temperature conditions calibration on the number of choices and the prompt length. Serving prompt, answer codes and adapter format are unchanged, so PACT is a drop-in replacement.
- 2) **A controlled, falsifiable evaluation.** We reproduce the published recipe inside one harness, add a control that shares PACT’s optimiser but not its loss terms, run every leave-one-out ablation, state in advance which observation would falsify each claim (Section V-E), and report paired significance tests rather than leaderboard deltas.
- 3) **An honest account of what the terms buy.** PACT does not raise accuracy over the published recipe. It does produce the lowest position bias and ordinal error of all runs, stabilises an aggressive optimiser that on its own is significantly worse, and makes calibration a cheap post-hoc step (Section VI); the ablations show which terms carry which effect and where the method does not help.

Fig. 1 summarises the method.

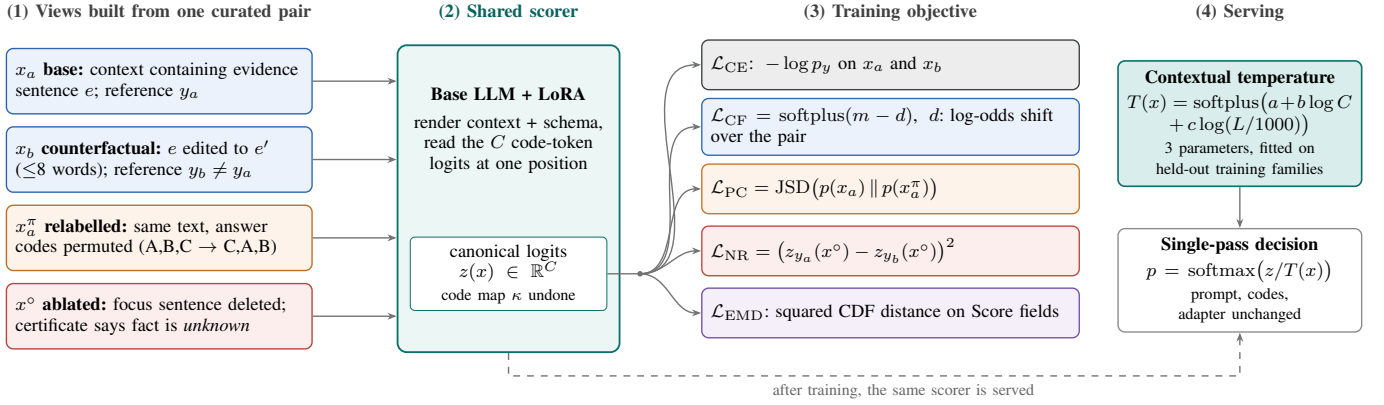


Fig. 1. **PACT at a glance.** (1) Each curated pair yields two labelled views (base, counterfactual), a copy of the base with its answer codes relabelled, and a certificate-verified ablated context with the decisive sentence removed. (2) All views share one forward pass per view through the LoRA-adapted model and are mapped to canonical answer order. (3) Four terms are added to cross-entropy, each using structure the released recipe discards. (4) A three-parameter temperature, fitted on training-side families, calibrates the served single-pass distribution. The colours of the views and loss terms are used consistently throughout the paper.

II. RELATED WORK

Serving typed decisions. Scoring candidate-token logits at a single position instead of generating and parsing structured text underlies the single-token decision interface described for this model family [1], [2]; MiniCheck [4] applies the same idea to one Boolean grounding decision. These works focus on the serving interface and data; we keep both fixed and change only the objective.

Counterfactual and contrastive supervision. Counterfactually augmented data [5] shows that minimal label-flipping edits reduce reliance on spurious features, and contrast sets [6] use the same construction for evaluation. Both usually feed edited examples to an unchanged objective. Teney et al. [7] supervise the input-gradient direction between counterfactual pairs; our margin instead supervises the *output* geometry of a pair and is exactly invariant to a shared offset, in the spirit of difference-in-differences estimation [8] and pairwise ranking losses [9]. Evidence removal is standard for *evaluating* rationales [10]; we use certified removals as a *training* constraint.

Option-order and label bias. LLM multiple-choice answers depend on option order and label identity [11]–[13], and on label priors more generally [14]. Debiasing is typically applied at inference by permuting and averaging or by re-normalising. We additionally train against the bias with a bounded Jensen–Shannon consistency loss [15], analogous to the consistency term of AugMix [16], which keeps single-pass serving—the reason this model family exists—sound.

Calibration. Temperature scaling [17] is the standard post-hoc method, with richer maps such as Dirichlet calibration [18]; LLM confidence is known to be miscalibrated in task-dependent ways [19], [20]. We keep the temperature-scaling objective but let the temperature depend on two observable prompt statistics, which preserves every argmax.

Ordinal targets and efficient tuning. Ordered labels benefit from losses that respect the order [21]; squared earth-mover losses [22] are one such choice, which we apply to rubric fields

whose expected level the serving path already reports. The adapter uses LoRA [3] with rank-stabilised scaling [23] and LoRA+ learning-rate ratios [24]; a QLoRA [25] configuration is provided for 24 GB GPUs.

III. PROBLEM SETTING

A. Single-token typed decisions

A field has allowed answers $v_1, \dots, v_C$ with $1 \leq C \leq 26$. The prompt renders the context, the schema and the requested field, and shows answer v_k under a code letter. With h the final-position hidden state and t_k the token of the k -th code, the model produces $z_k = \langle w_{t_k}, h \rangle$ and serves $p = \text{softmax}(z_1, \dots, z_C)$. Fields are of three kinds: *Choice* (an enumeration), *Boolean* (true/false) and *Score* (an ordered rubric with levels $0, \dots, C-1$). Two properties matter below. First, the map from a *meaning* to a *letter* is arbitrary, so any dependence of p on the letter is error. Second, since one position is scored, the decision is a linear read-out of one hidden state, which makes the geometry of z —differences between candidate logits—the natural object to supervise.

B. Contrastive curation with certificates

The training file contains 1,338 families of exactly two records: a base context and a counterfactual that differs in one edited evidence sentence, with the reference answer flipped and a byte-identical question and criteria block. Each record carries a certificate naming the two sentences that jointly determine the focus fact, their base and edited wording, and the verified statement that with either focus sentence removed the focus fact becomes *unknown*. The released recipe trains on the two records independently and does not construct the removed-evidence contexts, correctly noting that “missing evidence” has no reference label. Our observation is that the absence of a label is not the absence of information: the certificate establishes *indifference*, which is a constraint on the logit geometry rather than a target. Fig. 2 shows a real

pair from the training file and the three views PACT derives from it.

C. Notation

For an example x with C allowed answers, $z(x) \in \mathbb{R}^C$ denotes the candidate logits in *canonical* order (the order in which the schema declares the answers) and $p(x)$ the corresponding distribution. A pair is (x_a, x_b) with references $y_a \neq y_b$ from the same canonical set; x° is x_a with one focus evidence sentence deleted; π is a permutation of code positions and z^π the canonical logits obtained when the prompt is rendered under π .

IV. METHOD

A. Canonical choice space

Every loss and metric is computed after mapping code positions back to canonical answer indices. Per view, the encoder records a vector κ with κ_k the canonical index of the answer shown at code position k ; logits are scattered from code space into canonical space and unused positions are masked to $-\infty$. This makes the pair terms well defined when the two members are rendered under different permutations, and makes “the model changed its answer when the letters changed” a measurable event.

B. Counterfactual difference-in-differences margin

For a pair define

$$d(x_a, x_b) = [z_{y_a}(x_a) - z_{y_b}(x_a)] + [z_{y_b}(x_b) - z_{y_a}(x_b)], \quad (1)$$

i.e. the change of the $y_a:y_b$ log-odds between the two members, and penalise $\mathcal{L}_{\text{CF}} = \text{softplus}(m - d)$.

Proposition 1 (Offset invariance): For any $c \in \mathbb{R}^C$, replacing $z(x_a)$ by $z(x_a) + c$ and $z(x_b)$ by $z(x_b) + c$ leaves d unchanged. Moreover, if both members are correct under the binary restriction to $\{y_a, y_b\}$, then $d > 0$.

Proof: Each bracket in (1) is a difference of two logits of the same input, so $c_{y_a} - c_{y_b}$ enters the first bracket and $c_{y_b} - c_{y_a}$ the second; they cancel. Correctness of $x_a(x_b)$ under the restriction means the first (second) bracket is positive. ■

A per-pair offset c is exactly what a model learns when it acquires a prior over outcomes from the topic, the policy text or the family’s surface form. $\mathcal{L}_{\text{CF}}$ therefore cannot be reduced by such a prior, only by making the logit gap move in the right direction when the one edited fact moves. Cross-entropy lacks this property: a model that assigns 0.51 to y_a on the base and 0.49 on the counterfactual satisfies both cross-entropy terms nearly as well as a model with a decisive, evidence-driven gap. Because d couples both members, it can only be evaluated when both are in the same micro-batch (Section IV-G). We use $m = 2$ nats and a softplus hinge so that gradients vanish smoothly once the pair is separated.

C. Permutation consistency

For two code permutations π, π' of the same example we penalise

$$\mathcal{L}_{\text{PC}} = \text{JSD}(p^\pi \| p^{\pi'}) = \frac{1}{2} \text{KL}(p^\pi \| \bar{p}) + \frac{1}{2} \text{KL}(p^{\pi'} \| \bar{p}), \quad (2)$$

with $\bar{p} = \frac{1}{2}(p^\pi + p^{\pi'})$. JSD is bounded and symmetric, so no view is privileged as a teacher and a confidently wrong view cannot produce an unbounded gradient; the term is label-free. At inference the same invariance supports a K -view ensemble that averages canonical log-probabilities. We treat the gain from $K=1$ to $K=4$ as a measure of the bias that training failed to remove: a perfectly permutation-invariant model gains nothing.

D. Evidence-necessity regularisation

The certificate states that in x° the focus fact is unknown given all remaining text, and the decision rules that separate y_a from y_b take the focus fact as input. Hence the evidence in x° does not discriminate y_a from y_b —although it may still rule out other answers. We encode exactly that and nothing more:

$$\mathcal{L}_{\text{NR}} = \left(\text{relu}(|z_{y_a}(x^\circ) - z_{y_b}(x^\circ)| - \varepsilon) \right)^2. \quad (3)$$

The term is silent about $z_j(x^\circ)$ for $j \notin \{y_a, y_b\}$, about the entropy of $p(x^\circ)$, and about which of y_a, y_b is “correct”—claims the certificate does not support. To build x° we locate the focus sentence by certificate path, then exact leaf match, then substring, then the counterfactual wording, and delete the whole conversational turn if it carries nothing else. On the released training file this succeeds for all 2,676 records, verified by checking that the sentence no longer occurs anywhere in the serialised context. We use $\varepsilon = 0$.

E. Ordinal transport for rubric fields

Score fields expose levels $0, \dots, C-1$ and the serving code reports the expected level $\sum_j p_j$. Cross-entropy is invariant to the ordering of the levels, so nothing in training prefers a near miss to a far one. We add the squared earth-mover distance between predicted and reference CDFs,

$$\mathcal{L}_{\text{EMD}} = \sum_{j=0}^{C-1} \left(\sum_{i \leq j} p_i - \mathbf{1}[j \geq y] \right)^2, \quad (4)$$

which is differentiable and minimised by mass concentrated *near* y rather than merely *on* y .

F. Contextual temperature

The released model ships without calibration. A single global temperature has the wrong shape here: a two-choice Boolean and a six-choice enumeration do not share a miscalibration factor, and neither do a 300-token and a 1,100-token prompt. We fit

$$T(x) = \text{softplus}(a + b \log C + c \log(L/1000)), \quad (5)$$

with L the prompt length in tokens, by minimising NLL with L-BFGS on a *calibration split* of training-side families

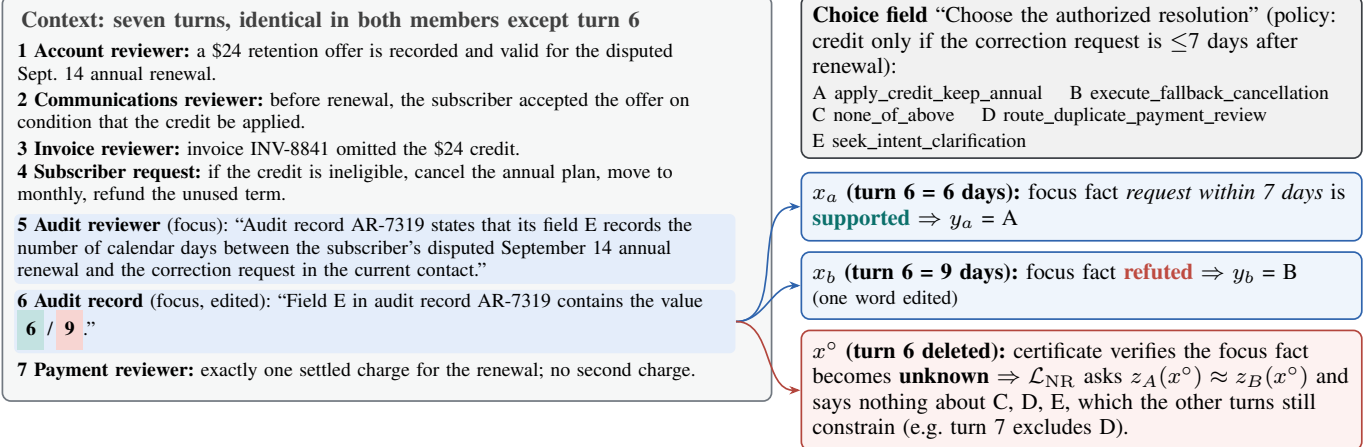


Fig. 2. **A curated contrastive pair from the training file** (family scale-diverse-013-002, commerce domain; turns 1–4 and 7 abridged, focus sentences verbatim). Changing one token of the second focus sentence (6→9 days) flips the certified focus fact and the reference answer from A to B; deleting that sentence makes the fact *unknown*. Cross-entropy sees x_a and x_b as two unrelated examples; PACT uses the pair ($\mathcal{L}_{CF}$), a relabelled copy ($\mathcal{L}_{PC}$) and the ablated context ($\mathcal{L}_{NR}$).

Algorithm 1 One PACT optimiser step

Require: groups $\mathcal{G}$ (pair (x_a, x_b)), permuted x_a^π, x_b^π , ablated x° ; ramp $r_t \in [0, 1]$

- 1: **for** each micro-batch $B \subset \mathcal{G}$ (length-bucketed) **do**
- 2: one forward pass over all views in B ; read code logits
- 3: scatter to canonical order with κ ; mask unused slots
- 4: $\ell \leftarrow \mathcal{L}_{CE} + r_t \lambda_{emd} \mathcal{L}_{EMD}$ ▷ labelled views
- 5: $\ell \leftarrow \ell + r_t (\lambda_{cf} \mathcal{L}_{CF} + \lambda_{pc} \mathcal{L}_{PC} + \lambda_{nr} \mathcal{L}_{NR})$
- 6: accumulate $\nabla \ell$
- 7: **end for**
- 8: clip global norm to 1; AdamW step on LoRA A (η) and B (8η)

never used to fit the adapter. With $b = c = 0$ this is standard temperature scaling [17]. Because $T(x) > 0$ rescales each item’s logits by a positive constant, it never changes an argmax, so accuracy and pair accuracy are untouched; it can, however, change the *ranking* of confidences across items and hence selective risk.

G. Objective, batching and cost

The full objective is

$$\mathcal{L} = \mathcal{L}_{CE} + \lambda_{cf} \mathcal{L}_{CF} + \lambda_{pc} \mathcal{L}_{PC} + \lambda_{nr} \mathcal{L}_{NR} + \lambda_{emd} \mathcal{L}_{EMD}, \quad (6)$$

with $(\lambda_{cf}, \lambda_{pc}, \lambda_{nr}, \lambda_{emd}) = (0.5, 0.5, 0.2, 0.3)$, chosen for scale comparability and *not* tuned on any evaluation data. The regularisers ramp in linearly over the first 10% of steps. Since $\mathcal{L}_{CF}$ needs both pair members and $\mathcal{L}_{PC}$ needs both views, the unit of batching is a *group*: one family with all of its views (Algorithm 1). Groups are shuffled, sorted by length inside buckets of 64 groups (prompt lengths vary by more than $3\times$, so padding dominates waste) and cut into micro-batches. Each pair contributes two primary, two permuted and one ablated view, i.e. $2.5\times$ the rows of the baseline.

TABLE I

DATA SPLITS. SPLITS ARE DISJOINT AT THE LEVEL OF SOURCE FAMILIES; EVERY RECORD BELONGS TO A TWO-RECORD CONTRASTIVE PAIR. THE HOLDOUT IS NEVER USED FOR SELECTION OR CALIBRATION.

Split	Fam.	Records	Pairs	Choice	Bool.	Score	Role
Training	30	2,334	1,167	766	796	772	train
Selection	2	116	58	26	32	58	select
Calibration	2	226	113	64	60	102	fit T
Holdout (frozen)	6	324	162	146	114	64	test only

V. EXPERIMENTAL SETUP

A. Data and splits

Training uses the released training file (2,676 records, 1,338 pairs, 34 source families, 10 domains). Four source families are held out from training: two form the *selection split* used for checkpoint selection and two the *calibration split* used to fit $T(x)$ (Table I). The frozen *holdout* (324 records, 162 pairs, 6 source families from 6 domains) is used only to report final numbers—never for selection, calibration or early stopping. Family-level disjointness and the file hashes of the dataset manifest are re-verified at load time. Holdout fields have $C \in \{2, \dots, 6\}$ allowed answers, and the longest rendered prompt has 1,098 tokens.

B. Model and training

The base model is Qwen3.5-9B [26] at the revision pinned by the published recipe, in BF16 with a 2,048-token limit. The adapter is LoRA of rank 16 ($\alpha=32$, dropout 0.05) on all attention and MLP projections of the language model—the modules the published recipe adapts—with rsLoRA scaling. We train with AdamW [27] at $\eta = 5 \times 10^{-5}$, LoRA+ ratio 8, cosine schedule with 10% warm-up, two epochs (584 steps) and an effective batch of 8 primary rows (4 pairs) per step, matching the published recipe. The selection split is scored

TABLE II

COMPARED RUNS. ALL SHARE DATA, PROMPT, ADAPTER FAMILY AND CHECKPOINT SELECTION. “OPT. A”: 1 EPOCH, LINEAR SCHEDULE, PLAIN LORA (PUBLISHED). “OPT. B”: 2 EPOCHS, COSINE, RSLOA + LORA+.

Run	Objective	Views [†]	Opt.	Seeds
Base model	– (adapter off)	–	–	–
Nimble recipe	$\mathcal{L}_{\text{CE}}$	2/0/0	A	3
CE-only	$\mathcal{L}_{\text{CE}}$	2/0/0	B	3
PACT	Eq. (6)	2/2/1	B	3
$-\mathcal{L}_{\text{CF}}, -\mathcal{L}_{\text{NR}}, -\mathcal{L}_{\text{EMD}}$	PACT minus one term	2/2/1	B	1
$-\mathcal{L}_{\text{PC}}$	PACT minus JSD	2/2*/1	B	1
–permutated views	PACT without $\mathcal{L}_{\text{PC}}$	2/0/1	B	1

[†]Primary / permuted / ablated views per pair. *Permuted views kept as plain cross-entropy augmentation.

every 146 steps and the checkpoint with the lowest selection NLL is kept. Jobs ran one per GPU on two 48 GB GPUs.

C. Compared runs

Table II lists the runs. Nimble recipe reproduces the published recipe inside our harness. CE-only shares *everything* with PACT except the four terms and the extra views, so PACT vs. CE-only isolates the objective, while Nimble recipe vs. CE-only isolates the optimisation changes. The three main arms use seeds 17, 18 and 19; each leave-one-out ablation uses seed 17 and is compared against PACT at the *same* seed (Section VI-D).

D. Metrics and statistics

We report accuracy; **pair accuracy**, the share of the 162 holdout pairs with *both* members correct; NLL; multi-class Brier score [28]; ECE over 15 equal-width bins [29]; the area under the confidence-ranked risk–coverage curve (AURC) [30]; expected-level MAE on Score fields; and, for $K=4$ decoding, the mean pairwise **total variation (TV)** between the canonical distributions of different code orderings and the **flip rate**, the share of items whose argmax changes across orderings. Because all runs are scored on the same items, we compare accuracies with the exact McNemar test [31] and a 10,000-resample paired bootstrap [32], per seed.

E. Pre-specified falsification criteria

The following were written before the runs, so the ablations read as tests rather than a search over cells.

- H1** *Pair margin.* Removing $\mathcal{L}_{\text{CF}}$ lowers pair accuracy by more than it lowers accuracy.
- H2** *Permutation consistency.* PACT has lower TV and flip rate than CE-only and gains less from $K=4$ ensembling.
- H3** *Necessity.* Without $\mathcal{L}_{\text{NR}}$ the model is more confident on ablated contexts and has no better pair accuracy.
- H4** *Ordinal transport.* Without $\mathcal{L}_{\text{EMD}}$, Score MAE is higher at similar Score accuracy.
- H5** *Calibration.* Holdout ECE falls after applying $T(x)$ without a loss of accuracy.

VI. RESULTS

A. Main comparison

Table III and Fig. 3 give the headline numbers, and Table IV the paired tests. Three findings stand out.

PACT matches, but does not beat, the published recipe in accuracy. PACT reaches $84.6 \pm 2.8\%$ against $85.2 \pm 2.1\%$ for Nimble recipe. At every seed the difference is inside the bootstrap interval and McNemar $p \geq 0.50$; at two seeds the accuracies are identical. Properly reproduced with checkpoint selection, the published recipe is a strong accuracy baseline, and it is also the best-calibrated run (ECE_T 5.4% vs. 7.9% for PACT).

The optimisation changes alone hurt; the PACT terms repair them. CE-only—PACT’s optimiser and schedule without the four terms—is the weakest and least stable fine-tuned arm ($82.2 \pm 6.0\%$, NLL 0.760, ECE 14.3%). Adding the terms raises mean accuracy by 2.4 pp, significantly so at seeds 18 and 19 ($p = 0.008$ and 0.035 ; seed 17 favours CE-only, $p = 0.099$), halves the seed-to-seed standard deviation ($6.0 \rightarrow 2.8$ pp) and lowers NLL by 26% and ECE by 25%. The terms thus act as regularisers that make a more aggressive optimiser safe, rather than as a source of accuracy beyond a well-tuned baseline.

Where PACT does lead, the lead matches the structure it encodes. PACT has the highest mean pair accuracy (73.3%, not significant given the 6 pp seed spread), the lowest Score MAE of all multi-seed arms (0.232 vs. 0.311 and 0.322), and the lowest position bias (Section VI-B). PACT clears the base model by 18–24 pp ($p < 0.001$ at every seed).

B. Position bias and permutation ensembling

Table V reports $K=4$ decoding. PACT has the lowest TV (5.6% vs. 6.9% for Nimble recipe, 7.4% for CE-only and 12.0% for the base model) and the lowest flip rate (9.8% vs. 13.8%, 12.1% and 23.5%). It is also the only arm that gains nothing from ensembling ($\Delta_K = -0.7$ pp vs. $+2.4$ pp for CE-only): its single pass already captures what the ensemble would recover, which is precisely the behaviour H2 predicts and the property that matters for single-pass serving.

Fig. 4 looks closer. The bias of the shipped model is concentrated: the median item has TV of 1.9% and 31% of items are below 1%, while a small tail of items carries TV above 30%. A local 4-bit sweep over $K \in \{1, \dots, 4\}$ (single seed; supplementary) shows the residual TV of PACT plateauing at 7.3–7.6% against 16.7–18.1% for the base model, $2.2\text{--}2.4\times$ lower at every K , and accuracy peaking at $K=2$ for both models. The non-monotone accuracy curve is not a quantisation artefact: the BF16 multi-seed results show the same pattern (84.6% at $K=1$, 83.8% at $K=4$).

C. Calibration

The contextual temperature lowers holdout ECE in all nine multi-seed runs (Fig. 5c; per-run values in Table X, Appendix A), on average from 10.8% to 7.9% for PACT, from 9.5% to 5.4% for Nimble recipe and from 14.3% to 10.3% for CE-only; accuracy is unchanged by construction.

TABLE III

HOLDOUT RESULTS, SINGLE-PASS DECODING ($K=1$), MEAN $\pm$ S.D. OVER SEEDS 17/18/19 (324 ITEMS, 162 PAIRS). ACCURACY, PAIR ACCURACY, ECE AND AURC IN %. ECE_T IS AFTER THE CONTEXTUAL TEMPERATURE $T(x)$ (NOT FITTED FOR THE BASE MODEL). **SHADED**: BEST MEAN PER COLUMN.

Run	Acc. $\uparrow$	Pair acc. $\uparrow$	NLL $\downarrow$	Brier $\downarrow$	ECE $\downarrow$	$ECE_T \downarrow$	AURC $\downarrow$	Score MAE $\downarrow$
Base model	66.4	35.8	1.094	0.549	24.6	—	—	0.571
Nimble recipe	85.2 $\pm$ 2.1	71.0 $\pm$ 3.9	0.449 $\pm$ 0.110	0.236 $\pm$ 0.042	9.5 $\pm$ 3.4	5.4 $\pm$ 0.8	3.41 $\pm$ 0.55	0.311 $\pm$ 0.030
CE-only	82.2 $\pm$ 6.0	70.2 $\pm$ 6.2	0.760 $\pm$ 0.296	0.305 $\pm$ 0.110	14.3 $\pm$ 6.4	10.3 $\pm$ 4.7	4.46 $\pm$ 2.94	0.322 $\pm$ 0.153
PACT (ours)	84.6 $\pm$ 2.8	73.3 $\pm$ 6.2	0.560 $\pm$ 0.088	0.251 $\pm$ 0.026	10.8 $\pm$ 1.7	7.9 $\pm$ 1.1	3.95 $\pm$ 1.03	0.232 $\pm$ 0.120

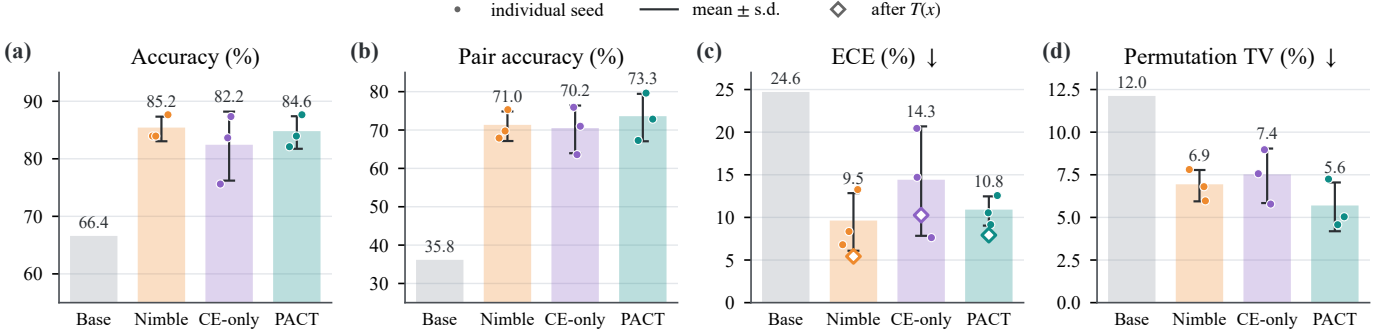


Fig. 3. Holdout comparison of the three multi-seed arms and the base model. Bars: mean; dots: individual seeds; whiskers: ± 1 s.d. (a–c) single-pass decoding; in (c) hollow diamonds give ECE after the contextual temperature. (d) Permutation TV under $K=4$ relabellings (lower is less position bias).

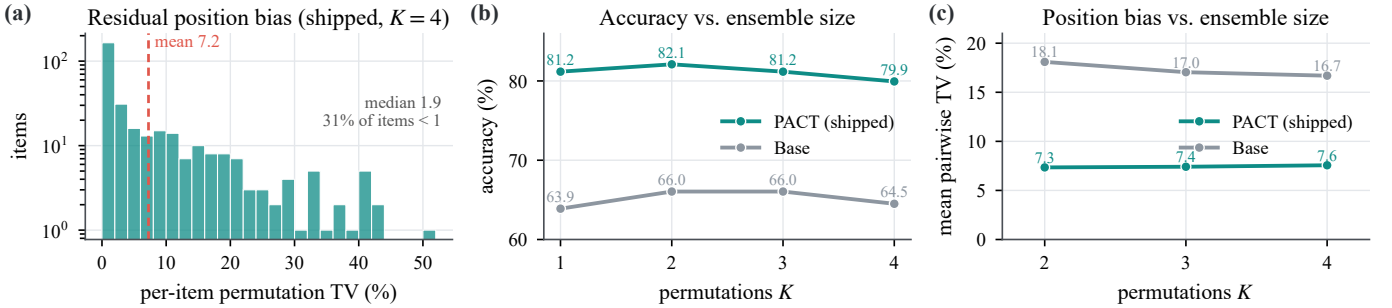


Fig. 4. Position bias. (a) Distribution of per-item permutation TV for the shipped PACT model under $K=4$ (log scale). (b, c) Local 4-bit sweep over the ensemble size K for the shipped model and the base model: accuracy (b) and mean pairwise TV between orderings (c).

For the shipped model the fitted slopes are negative for both the number of choices ($b = -0.42$) and the prompt length ($c = -0.48$), i.e. fields with many choices and long prompts are *under-confident* relative to short binary ones, which a single scalar temperature cannot express (Fig. 6a).

Three caveats are reported rather than omitted. First, the *shape* of the fitted temperature is not stable across runs (Fig. 6b): over the nine multi-seed runs the slope on $\log C$ ranges from -1.60 to $+2.73$ and the slope on $\log(L/1000)$ from -3.49 to $+3.58$. With 226 calibration records the two slopes are weakly identified; what transfers to the holdout is most plausibly the overall *level* of the temperature, so a scalar temperature is a reasonable default and the contextual form should be regularised or fitted on a larger split. Second, on the calibration split of the shipped seed the fit leaves ECE essentially unchanged ($5.5\% \rightarrow 5.7\%$); the gain appears only on the holdout. Third, because $T(x)$ re-ranks confidences across items, it slightly worsens selective risk (AURC $3.95 \rightarrow 4.21$ for

PACT; Fig. 5b).

D. Seed-matched ablations

Table VI compares each leave-one-out variant with PACT *at the same seed*. Two caveats frame the reading. Each row is a single run, and the seed-17 PACT reference is itself a relatively weak draw (84.0% accuracy vs. 84.6% mean, and the highest ECE of its three seeds), so colours indicate directions, not significance. With that in mind, the pattern is consistent:

- **No single term raises raw accuracy.** Removing $\mathcal{L}_{CF}$ or $\mathcal{L}_{EMD}$ leaves accuracy unchanged (84.3% vs. 84.0%); removing $\mathcal{L}_{PC}$ even raises it (87.7%), as does removing all four terms (CE-only, 87.3% at this seed, though not at the other two). Every removal also improves uncalibrated NLL and ECE, i.e. at this seed the regularisers trade calibration of the raw logits for other properties—a cost the post-hoc temperature largely absorbs (Section VI-C).
- **The robustness properties are carried by the terms.** Removing $\mathcal{L}_{NR}$, $\mathcal{L}_{EMD}$, the permuted views or all four

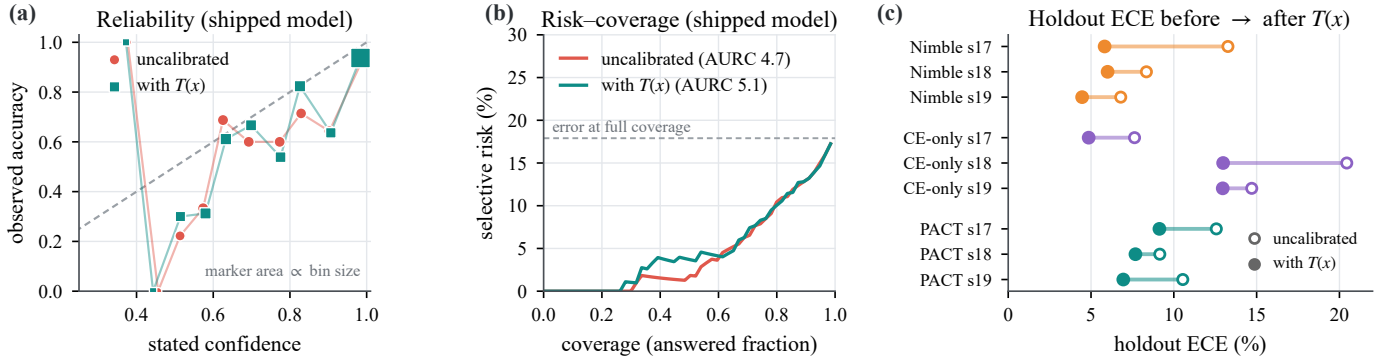


Fig. 5. Calibration. (a) Reliability diagram of the shipped model before and after $T(x)$; marker area grows with bin size. (b) Risk-coverage curves, items ranked by confidence. (c) Holdout ECE before (hollow) and after (filled) $T(x)$ for every run and seed.

TABLE IV
PAIRED TESTS ON HOLDOUT ACCURACY (%), SAME 324 ITEMS. b/c : ITEMS ONLY PACT / ONLY THE CONTROL GETS RIGHT; EXACT TWO-SIDED McNemar p ; 95% CI OF THE ACCURACY DIFFERENCE FROM A 10,000-RESAMPLE PAIRED BOOTSTRAP (PP). **HIGHLIGHTED**: $p < 0.05$. [†]BASE-MODEL PREDICTIONS ARE FROM A LOCAL 4-BIT RE-EVALUATION (ACC. 63.9%).

Seed	PACT	Ctrl.	Δ	b/c	p	95% CI
<i>PACT vs. Nimble recipe</i>						
17	84.0	84.0	0.0	19/19	1.000	[−3.7, +3.7]
18	82.1	84.0	−1.9	25/31	0.504	[−6.5, +2.8]
19	87.7	87.7	0.0	27/27	1.000	[−4.6, +4.3]
<i>PACT vs. CE-only</i>						
17	84.0	87.3	−3.4	13/24	0.099	[−7.1, +0.3]
18	82.1	75.6	+6.5	39/18	0.008	[+1.9, +11.1]
19	87.7	83.6	+4.0	23/10	0.035	[+0.6, +7.4]
<i>PACT vs. Base model[†]</i>						
17	84.0	63.9	+20.1	92/27	<0.001	[+13.9, +26.2]
18	82.1	63.9	+18.2	86/27	<0.001	[+12.0, +24.4]
19	87.7	63.9	+23.8	98/21	<0.001	[+17.9, +29.9]

TABLE V
PERMUTATION-ENSEMBLED DECODING ($K=4$), %. TV: MEAN PAIRWISE TOTAL VARIATION BETWEEN THE K CANONICAL DISTRIBUTIONS; FLIP: SHARE OF ITEMS WHOSE ARGMAX CHANGES ACROSS ORDERINGS. Δ_K : ACCURACY GAIN OF $K=4$ OVER $K=1$ (PP); A POSITION-INVARIANT MODEL SHOULD GAIN NOTHING.

Run	Acc.	Pair	ECE	TV↓	Flip↓	Δ_K
Base model	67.6	38.3	21.4	12.0	23.5	+1.2
Nimble recipe	85.7 ± 2.6	72.4 ± 4.4	4.7 ± 1.0	6.9 ± 0.9	13.8 ± 1.8	+0.5
CE-only	84.6 ± 4.6	73.3 ± 6.8	8.2 ± 3.4	7.4 ± 1.6	12.1 ± 3.8	+2.4
PACT (ours)	83.8 ± 2.6	72.4 ± 5.5	8.6 ± 1.6	5.6 ± 1.4	9.8 ± 2.1	−0.7

terms raises the flip rate (8.0% $\rightarrow$ 13.0–17.6%) and TV; removing $\mathcal{L}_{CF}$, $\mathcal{L}_{NR}$, $\mathcal{L}_{EMD}$ or the permuted views raises Score MAE (0.229 $\rightarrow$ 0.282–0.356).

- **The necessity term is the most load-bearing.** Removing $\mathcal{L}_{NR}$ degrades accuracy (−3.1 pp), pair accuracy (−8.0 pp), AUC (4.44 $\rightarrow$ 6.63), Score MAE and position bias (TV 5.0 $\rightarrow$ 11.7%) at once.

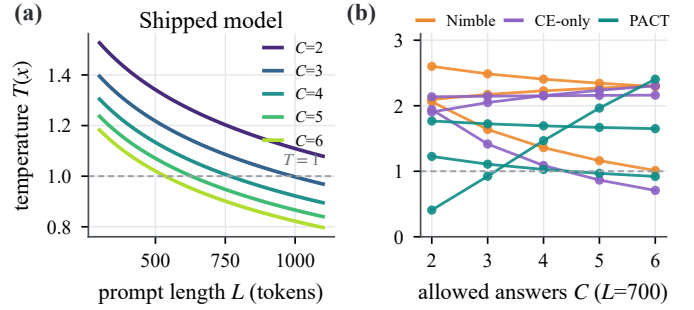


Fig. 6. Fitted contextual temperature. (a) Shipped model: $T(x)$ against prompt length for each number of allowed answers C in the holdout. (b) T against C at $L=700$ for every multi-seed run: for most runs $T > 1$ (raw logits are over-confident), but the slopes differ in sign between seeds.

E. Where the differences are

Pooled accuracy hides a structured, seed-consistent pattern (Fig. 8). Relative to Nimble recipe, PACT is better on Score fields at all three seeds (+6.2 to +15.6 pp) and worse on Choice (−3.4 to −4.8 pp) and Boolean fields (−0.9 to −4.4 pp) at all three seeds; per-kind accuracies are in Table IX (Appendix A). The Score advantage is what the ordinal term targets. By domain, PACT is better on *public services* and worse on *education* and *media* at every seed. We had no pre-specified hypothesis about domains and report this as an observation for future work: the gains of a structure-derived objective are heterogeneous, and a pooled accuracy delta near zero can conceal offsetting per-kind effects of 5–15 pp.

The same split appears in calibration (Fig. 7c). Averaged over seeds, PACT halves the ECE of Score fields relative to Nimble recipe (10.8% vs. 21.2%; 10.1% vs. 16.6% after $T(x)$) but is worse on Choice fields (16.0% vs. 10.3%), which is where its overall calibration deficit comes from. (Per-kind ECE is computed on 64–146 items and is biased upwards relative to the pooled value.)

How pairs fail. Fig. 7b breaks the shipped model’s 162 holdout pairs down by outcome. Both members are right in 67% of pairs. When a pair is only half right, it is usually the counterfactual that is missed (35 pairs vs. 13 in which only

TABLE VI

SEED-MATCHED LEAVE-ONE-OUT ABLATIONS (SEED 17 FOR EVERY ROW; SINGLE-PASS EXCEPT THE LAST TWO COLUMNS, WHICH USE $K=4$). CELLS ARE COLOURED BY THE CHANGE RELATIVE TO THE FULL METHOD IN THE FIRST ROW (SEED 17): REMOVING THE TERM CLEARLY HURTS, SLIGHTLY HURTS, SLIGHTLY HELPS, CLEARLY HELPS; UNCOLOURED: CHANGE BELOW A QUARTER OF THE “CLEAR” THRESHOLD (2 PP FOR ACCURACY-TYPE METRICS). ONE SEED PER ROW: READ COLOURS AS DIRECTIONS, NOT AS SIGNIFICANCE.

Variant (seed 17)	Acc. $\uparrow$	Pair acc. $\uparrow$	NLL $\downarrow$	ECE $\downarrow$	AURC $\downarrow$	Score MAE $\downarrow$	TV $K=4$ $\downarrow$	Flip $K=4$ $\downarrow$
PACT (ours)	84.0	72.8	0.659	12.6	4.44	0.229	5.0	8.0
$-\mathcal{L}_{CF}$	84.3	71.6	0.491	11.1	2.74	0.284	5.3	9.6
$-\mathcal{L}_{PC}$	87.7	79.6	0.513	7.5	2.82	0.138	4.8	9.3
$-\mathcal{L}_{NR}$	80.9	64.8	0.545	10.3	6.63	0.344	11.7	17.6
$-\mathcal{L}_{EMD}$	84.3	70.4	0.388	5.8	3.43	0.282	8.9	15.7
—permuted views	84.9	74.1	0.510	8.8	3.51	0.356	6.6	13.0
CE-only (all four off)	87.3	75.9	0.454	7.6	2.66	0.182	9.0	16.4

TABLE VII

PRE-SPECIFIED HYPOTHESES (SECTION V-E) AGAINST THE EVIDENCE. “SEED-MATCHED” VALUES COMPARE AGAINST PACT AT SEED 17; MULTI-SEED VALUES ARE MEANS OVER THREE SEEDS.

H	Prediction	Evidence	Verdict
H1	$-\mathcal{L}_{CF}$ loses more pair accuracy than accuracy	Seed-matched, $K=1$: accuracy +0.3 pp, pair accuracy -1.2 pp (direction as predicted); under $K=4$ the direction reverses (+1.5 / +2.5 pp). Mechanism confirmed: d rises from -0.3 to > 10 nats (Fig. 9).	INCONCLUSIVE
H2	Lower TV/flip than CE-only; smaller $K=4$ gain	TV 5.6 vs. 7.4%, flip 9.8 vs. 12.1%, Δ_K -0.7 vs. $+2.4$ pp; also below the Nimble recipe. Seed-matched: dropping the JSD term raises flips $8.0 \rightarrow 9.3\%$, dropping all permuted views $8.0 \rightarrow 13.0\%$.	SUPPORTED
H3	$-\mathcal{L}_{NR}$ more confident on ablated contexts; no better pair accuracy	Confidence on ablated contexts was not logged for $-\mathcal{L}_{NR}$ (untested half). Pair accuracy -8.0 pp, accuracy -3.1 pp, TV $5.0 \rightarrow 11.7\%$; with $\mathcal{L}_{NR}$ the ablated-context gap shrinks from 3.4 to < 0.1 nats.	PARTLY TESTED
H4	$-\mathcal{L}_{EMD}$ raises Score MAE at similar Score accuracy	Score MAE $0.229 \rightarrow 0.282$, but Score accuracy is <i>not</i> similar ($79.7 \rightarrow 84.4\%$). Multi-seed: lowest Score MAE of all arms; Score accuracy above the Nimble recipe at all three seeds.	PARTLY SUPPORTED
H5	ECE falls under $T(x)$, accuracy kept	Holdout ECE lower in 9/9 runs (PACT $10.8 \rightarrow 7.9\%$); accuracy unchanged by construction; AURC slightly worse ($3.95 \rightarrow 4.21$).	SUPPORTED

the base is missed; for Boolean fields all 11 half-right pairs miss the counterfactual). Most strikingly, in 39 pairs (24%; 33% for Choice fields) the model gives the *same* answer to both members, i.e. it ignores the edited fact altogether—74% of all imperfect pairs. This is exactly the failure $\mathcal{L}_{CF}$ targets, and it remains the dominant error mode after training: the mean margin on training pairs ends far above m (Fig. 9b), but the behaviour does not fully transfer to new source families. Errors are also not always hesitant (Fig. 7a): the median wrong answer has confidence 0.74 against 0.995 for right answers, yet 16 of the 58 errors (28%) are made with confidence ≥ 0.9 , against 56 of 117 (48%) for the base model.

F. Training dynamics

Fig. 9 shows that the terms are optimised as intended. The pair margin d starts negative (-0.3 nats: the base model moves the log-odds the wrong way), first exceeds the margin $m = 2$ at step 20 and, after early oscillation, ends near 10–13 nats. The logit gap on ablated contexts falls from 3.4 to below 0.1 nats, i.e. the model learns to be indifferent between y_a and y_b exactly where the certificate says it should. $\mathcal{L}_{NR}$ is also the noisiest term, with isolated spikes when a batch contains an ablated context the model still separates strongly.

G. Hypothesis scorecard

Table VII confronts each pre-specified criterion with the evidence. Two hypotheses are supported, two partly, and one is inconclusive; none is contradicted outright, but H1—the motivating idea of the paper—does not receive clean support from a single-seed ablation.

VII. DISCUSSION

What the evidence supports. The results do not support the proposition that structure-derived terms raise accuracy over a well-reproduced cross-entropy baseline. They do support three narrower claims: (i) the terms reduce answer-code position bias and ordinal error, the two failure modes they were designed around; (ii) they stabilise an optimiser that is otherwise significantly worse and highly seed-sensitive; and (iii) post-hoc calibration of this model family is cheap and lowers ECE in every run, although only the overall temperature level, not its contextual shape, is reliably estimated. For a practitioner who serves single-pass decisions over rubric fields, or who cannot afford K forward passes to average out position bias, these are the properties that matter; for a practitioner who only needs pooled accuracy, the published recipe is sufficient and about five times cheaper to train.

What remains open. The dominant holdout failure is still the one the pair margin was designed for: in a quarter of the

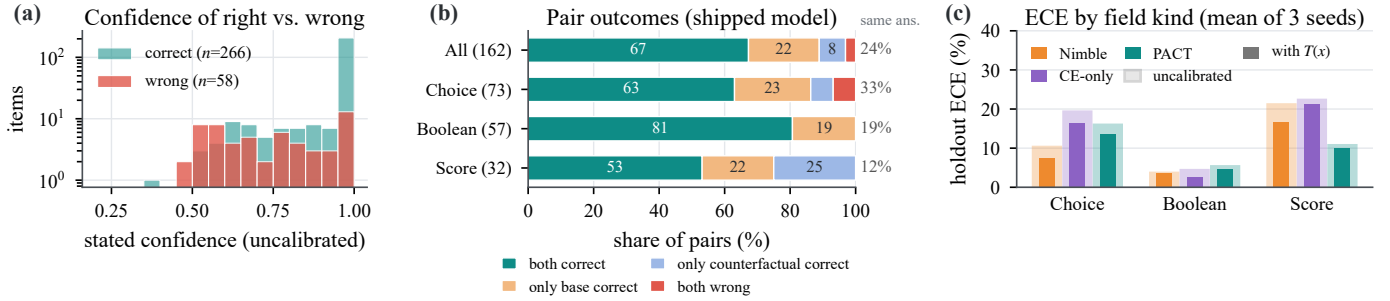


Fig. 7. Error analysis on the holdout. (a) Uncalibrated confidence of correct and wrong answers of the shipped model (log scale). (b) Outcomes of the 162 contrastive pairs by field kind (numbers: % of pairs; right: share of pairs where both members receive the same answer, which guarantees at least one error). (c) ECE by field kind, mean over three seeds: wide light bars uncalibrated, narrow dark bars after $T(x)$.

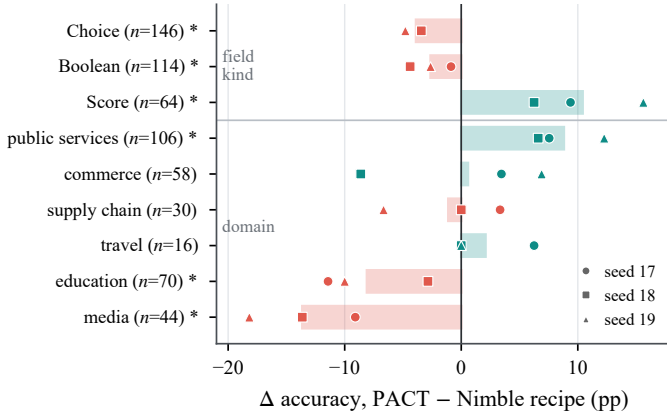


Fig. 8. Per-kind and per-domain accuracy difference, PACT minus Nimble recipe, on the holdout. Markers: seeds; bars: mean. *Same sign at all three seeds.

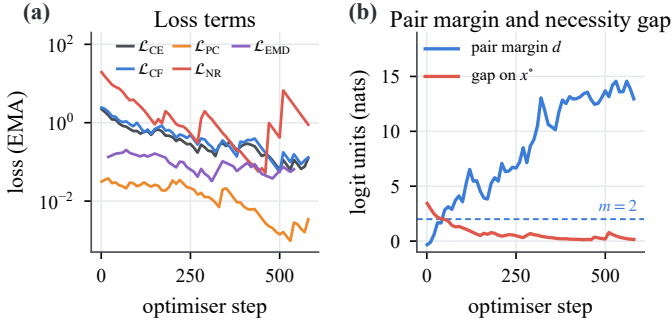


Fig. 9. Training dynamics of the shipped run (EMA-smoothed). (a) Loss terms. (b) Mean pair margin d of Eq. (1) (dashed: $m = 2$) and mean gap $|z_{y_a} - z_{y_b}|$ on ablated contexts x^* .

pairs the model answers both members identically, ignoring the edited fact (Section VI-E). The mean margin is met on training pairs but the behaviour does not fully transfer to unseen source families, which points to margin schedules, harder negative edits or pair-level data augmentation as the most promising next steps.

Cost. Each pair contributes $2.5\times$ the rows of the baseline (Table VIII). Measured wall-clock per run is 1.37 h for PACT against 0.50 h for CE-only and 0.26 h for the one-epoch

TABLE VIII
TRAINING COST PER RUN (ONE 48 GB GPU PER JOB): OPTIMISER STEPS, TRAINING VIEWS PER PAIR, MEAN WALL-CLOCK TIME, PEAK GPU MEMORY, AND MEAN HOLDOUT ACCURACY (%). SERVING COST IS IDENTICAL FOR ALL RUNS.

Run	Steps	Views/pair	Hours	Peak GiB	Acc.
Nimble recipe	292	2.0	0.26	21.8	85.2
CE-only	584	2.0	0.50	21.8	82.2
PACT	584	5.0	1.37	28.8	84.6
$-\mathcal{L}_{CF}$	584	5.0	1.37	28.7	84.3
$-\mathcal{L}_{PC}$	584	5.0	1.36	28.7	87.7
$-\mathcal{L}_{NR}$	584	4.0	1.06	26.4	80.9
$-\mathcal{L}_{EMD}$	584	5.0	1.36	28.7	84.3
—permuted views	584	3.0	0.76	24.0	84.9

Nimble recipe, and peak GPU memory is 28.8 GiB against 21.8 GiB. Cost scales with the number of views rather than with the loss terms: dropping $\mathcal{L}_{PC}$ alone saves nothing, while dropping the permuted views saves 45% of the time. Serving cost is unchanged.

Selection noise. Checkpoint selection is itself a source of variance comparable to the effects studied here (Fig. 10). The shipped model is the PACT seed with the lowest selection-split NLL (seed 18), yet it has the lowest holdout accuracy of the three PACT seeds (82.1%). More strikingly, the checkpoint with the lowest selection NLL of *all* nine runs (CE-only seed 18: NLL 0.112, 97.4% selection accuracy) has the *lowest* holdout accuracy of all fine-tuned runs (75.6%). With only two source families and 116 records, the selection split rewards fitting those families rather than transfer, which is one more reason to report seeds individually and to test differences in pairs.

VIII. LIMITATIONS

The holdout contains 324 synthetic, model-labelled items from six source families. It is narrow, and its labels come from the same family of models as the training labels, so shared errors are possible and undetectable from inside the benchmark. $\mathcal{L}_{NR}$ and $\mathcal{L}_{CF}$ trust the certificate: a wrong necessity check teaches indifference where the evidence actually decides. We apply the necessity term to one ablated view per pair to bound cost. The ablations use one seed each, so per-term claims are

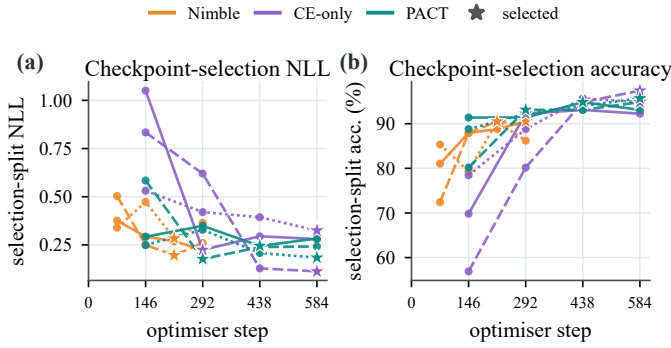


Fig. 10. Selection-split trajectories of the three multi-seed arms (line style: seed 17 solid, 18 dashed, 19 dotted; stars: selected checkpoints). CE-only converges slowly and reaches the best selection scores, which do not carry over to the holdout (Table IV).

directional; the significance tests cover only the three multi-seed arms. The loss weights were fixed a priori and not tuned, so the negative accuracy findings are for this setting, not for every weighting. All claims concern one base model at one scale with one adapter rank.

IX. CONCLUSION

Contrastive curation with certificates contains supervision that cross-entropy cannot express: the offset-free effect of an edited fact, the arbitrariness of answer codes, the indifference certified by evidence removal, and the order of rubric levels. PACT turns each into a training term without changing the serving contract. Evaluated under pre-specified falsification criteria, the terms do not make the model more accurate than a well-reproduced baseline, but they make it less sensitive to answer-code order, better on ordinal fields and more stable across seeds, and a three-parameter temperature makes its probabilities usable. We hope the protocol—a reproduced baseline, a matched-optimiser control, seed-matched ablations and paired tests—is as useful as the method.

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TABLE IX
HOLDOUT ACCURACY (%) BY FIELD KIND, SINGLE-PASS DECODING
(CHOICE $n=146$, BOOLEAN $n=114$, SCORE $n=64$), AND
EXPECTED-LEVEL MAE ON SCORE FIELDS.

Run	Seed	Choice	Boolean	Score	Score MAE
Base model	–	59.6	84.2	50.0	0.571
Nimble recipe	17	80.8	95.6	70.3	0.326
Nimble recipe	18	81.5	94.7	70.3	0.331
Nimble recipe	19	84.9	97.4	76.6	0.277
CE-only	17	78.8	97.4	89.1	0.182
CE-only	18	68.5	94.7	57.8	0.486
CE-only	19	78.1	94.7	76.6	0.299
PACT	17	77.4	94.7	79.7	0.229
PACT	18	78.1	90.4	76.6	0.353
PACT	19	80.1	94.7	92.2	0.114
$-\mathcal{L}_{CF}$	17	79.5	95.6	75.0	0.284
$-\mathcal{L}_{PC}$	17	80.1	94.7	92.2	0.138
$-\mathcal{L}_{NR}$	17	76.7	93.9	67.2	0.344
$-\mathcal{L}_{EMD}$	17	75.3	95.6	84.4	0.282
–permuted views	17	83.6	94.7	70.3	0.356

TABLE X
CONTEXTUAL TEMPERATURE PER RUN: MEAN [MIN, MAX] OF $T(x)$ ON
THE CALIBRATION SPLIT, AND ECE (%) BEFORE/AFTER ON THE
CALIBRATION SPLIT (FIT) AND ON THE HOLDOUT (TEST). * SHIPPED
MODEL. **Red**: $T(x)$ RAISED ECE.

Run	Seed	$T(x)$	Calib. split		Holdout	
			pre	post	pre	post
Nimble recipe	17	1.85 [1.10, 2.89]	14.4	6.5	13.3	5.8
Nimble recipe	18	1.54 [0.99, 2.00]	11.4	8.3	8.3	6.0
Nimble recipe	19	1.85 [1.28, 2.41]	16.1	9.1	6.8	4.5
CE-only	17	2.01 [1.78, 2.23]	15.5	6.0	7.6	4.9
CE-only	18	2.12 [1.95, 2.32]	15.7	4.5	20.4	13.0
CE-only	19	1.56 [0.70, 2.45]	8.1	7.6	14.7	13.0
PACT	17	1.56 [0.69, 2.69]	12.8	5.7	12.6	9.1
PACT	18*	1.16 [0.92, 1.40]	5.5	5.7	9.2	7.7
PACT	19	1.66 [1.61, 1.76]	12.5	6.0	10.5	6.9

APPENDIX A ADDITIONAL TABLES

Table IX breaks accuracy down by field kind for every run and seed; Table X gives the fitted temperatures and ECE before and after calibration on both the calibration split and the holdout.

APPENDIX B IMPLEMENTATION AND REPRODUCIBILITY

All code, configurations, data splits, trained adapters, run records and the scripts that regenerate every figure and table of this paper are released at <https://github.com/BennyLinntu/PACT-Pairwise-Anchored-Calibrated-Tuning-for-Single-Token-Typed-Decisions>. The training driver records, per run, the resolved configuration, the data audit and file hashes, the view-cache fingerprint, library versions, device information, the training plan, the full step log, the selection-split history, the fitted calibrator and every per-item holdout prediction with

TABLE XI
HYPER-PARAMETERS OF PACT (READ FROM THE RESOLVED RUN
CONFIGURATION). THE NIMBLE RECIPE DIFFERS ONLY IN: 1 EPOCH,
LINEAR SCHEDULE, NO RSLoRA, LoRA+ RATIO 1, NO EXTRA VIEWS OR
TERMS.

Setting	Value
Base model	Qwen3.5-9B, BF16, SDPA attention
Max. prompt length	2,048 tokens (longest seen: 1,098)
LoRA	rank 16, $\alpha=32$, dropout 0.05, rsLoRA
LoRA targets	q,k,v,o,gate,up,down (language model)
Optimiser	AdamW, lr 5×10^{-5} , weight decay 0
LoRA+ ratio	8 (lr of B / lr of A)
Schedule	cosine, warm-up 10%, 2 epochs
Batch	2 pairs $\times$ 2 accumulation = 8 primary rows
Gradient clipping	global norm 1.0
Loss weights	$\lambda_{cf}=0.5$, $\lambda_{pc}=0.5$, $\lambda_{nr}=0.2$, $\lambda_{emd}=0.3$
Margins	$m=2$, $\epsilon=0$; ramp 10% of steps
Length bucketing	64 groups per bucket
Selection	every 25% of training, lowest selection NLL
Evaluation	$K=4$ permutations, ECE with 15 bins
Calibration	contextual $T(x)$, L-BFGS, 120 iterations
Seeds	17, 18, 19 (ablations: 17)

its probabilities. Adapters are saved with the recipe’s `schema_config.json` contract, including the SHA-256 of the scoring-prompt implementation, so that a checkpoint is rejected if the prompt code it was trained against has changed. All figures and data tables in this paper are regenerated from these records by a single script. A 4-bit NF4 configuration for 24GB GPUs is provided but its results are not mixed with the BF16 numbers reported here; the only 4-bit numbers in the paper are the supplementary K -sweep and the base-model predictions used for the paired tests. Table XI lists every hyper-parameter; none was tuned on the holdout.

APPENDIX C APPLIED DEMONSTRATION: SEQUENTIAL DECISIONS IN TETRIS

Everything above evaluates the adapter on a static document plus a schema question. As a qualitative check of how general the serving interface is, we reused it unmodified for a task with no document at all: choosing where to place falling pieces in a from-scratch Tetris implementation. At every piece the game engine—not the model—enumerates the legal placements and computes their consequences two plies deep (rows cleared, holes, stack height and bumpiness, now and after the best placement of the visible next piece; standard Tetris features [33]). Each option becomes one choice description in an ordinary schema call; the model reads one token and the highest-logit option is played.

Policy-gradient adaptation. Unmodified, the adapter does not reliably choose the option the two-ply search ranks best: it was trained to classify documents, not to trade off sequential consequences. We therefore further fine-tuned the same LoRA adapter with single-step REINFORCE [34] (a contextual bandit over candidate placements), using the two-ply score of the sampled placement as reward and the mean score over

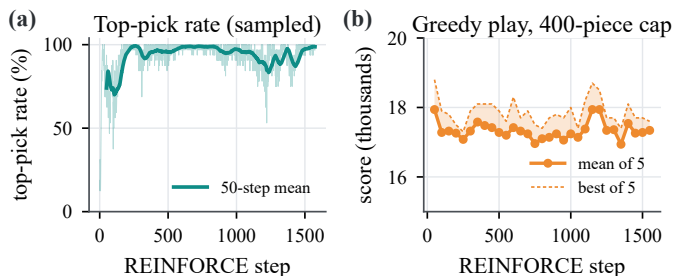


Fig. 11. REINFORCE on Tetris placement choices. (a) Share of sampled moves that match the two-ply search’s top-ranked option (faint: per update; bold: 50-step mean). (b) Greedy evaluation every 50 steps, 5 games capped at 400 pieces (line: mean; dashed: best game).

that state’s candidates as a per-state baseline, with an entropy bonus of 0.01, learning rate 10^{-5} and 32 moves per update on 2×40 GB GPUs. Prompt construction is shared between training and serving, so there is no train/serve skew.

Results. The sampled top-pick rate—agreement with the search’s best-ranked option—rose from 19% at the first update to 87.5% by step 16, and its 100-step average stayed between 88% and 99% from step 200 to the end of the 1,574-step run (Fig. 11a). The training game itself never topped out after step 14, clearing 19,975 lines over roughly 50,000 pieces. Greedy evaluation with a 400-piece cap (Fig. 11b) is flat at a mean of 17,000–18,000 points and about 158 lines, close to the ceiling of 160 lines that 400 pieces permit, so this protocol saturates and cannot show improvement; one uncapped greedy game at the final checkpoint reached 1,500 pieces and 598 lines (score 65,700) before it was stopped.

Live play in the served app. The repository (Appendix B) includes a three-minute recording at $2 \times$ speed (`media/pact_rl_tetris_2x.mp4`) of the final checkpoint driving the unmodified browser app through the ordinary serving path, in 4-bit on a single 16 GB consumer GPU. In six minutes of real play the model cleared more than 130 lines in one continuous game and reached level 13 without topping out, although gravity accelerates with every level and each move requires a fresh forward pass.

Reading this appendix. This is a single run without seeds, held-out repetitions or a greedy pre-training baseline, and the policy learns to follow a ranking that the engine already computes. It demonstrates only that the single-token interface is a general typed-choice interface and that the same lightweight adapter can be re-targeted to a new decision task; it carries no evidence about the holdout claims above.